THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Who proposed that all animals are made of cells (1839)? 1

(a) Theodor Schwann **(b)** Robert Hooke **(c)** Golgi **(d)** Rudolf Virchow

2. The total magnification of a microscope with a 10X eyepiece and a 40X objective is: 1

(a) 100X **(b)** 400X **(c)** 4X **(d)** 50X

3. The movement of water across a selectively permeable membrane is called: 1

(a) transpiration **(b)** diffusion of gases
(c) photosynthesis **(d)** osmosis

4. The fluid mosaic model describes the structure of: 1

(a) the cell membrane **(b)** the nucleus
(c) mitochondria **(d)** the cell wall

5. The control centre of a eukaryotic cell is the: 1

(a) Golgi apparatus **(b)** nucleus **(c)** ribosome **(d)** vacuole

6. Which cell does NOT have a nucleus when mature? 1

(a) human red blood cell **(b)** human nerve cell
(c) cheek cell **(d)** onion peel cell

7. The organelle that packages and dispatches materials in vesicles is the: 1

(a) Golgi apparatus **(b)** ribosome **(c)** chloroplast **(d)** nucleus

8. The plastids that give colour to flowers and fruits are: 1

(a) chloroplasts **(b)** lysosomes **(c)** chromoplasts **(d)** leucoplasts

9. Cell division in which one parent cell gives two identical daughter cells is: 1

(a) fertilisation **(b)** mitosis **(c)** osmosis **(d)** meiosis

10. Growing cells outside the body in a nutrient medium is known as: 1

(a) cell cycle **(b)** cell theory **(c)** cell culture **(d)** cell wall

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the function of the cell membrane? 2

12. Define osmosis. 2

13. Differentiate between diffusion and osmosis. 2

14. Why does a plant cell not burst in pure water while an animal cell may? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare prokaryotic and eukaryotic cells in three points. 3

16. An onion peel cell is estimated to be 200 µm. If 25 cells span the diameter of the field of view, find the diameter of the field. 3

17. What is the function of the endoplasmic reticulum? Distinguish RER and SER. 3

OR

What is the role of the Golgi apparatus? 3

18. Why are lysosomes called ‘suicide bags’? 3

OR

Explain how mitochondria and plastids support the idea that they evolved from bacteria. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Cell X swells and cell Y shrinks when placed in two solutions. 5

(i) What is the direction of water movement in each?
(ii) Which solution is more concentrated?
(iii) If X is a plant cell, why does it not burst?
(iv) What term describes the shrinking of the inner content of a plant cell like Y?

OR

(i) Draw and label a plant cell. 5
(ii) State two features that distinguish it from an animal cell.
(iii) Name the organelle involved in photosynthesis.

20. (i) What is the role of the nucleus? 5
(ii) Describe chromatin, chromosomes and genes.
(iii) Why can mature RBCs not divide?

OR

Case study: A student is to make a table of the organelles: mitochondria, ER, Golgi, lysosomes, ribosomes. 5

(i) Give the function of each.
(ii) Which two have their own DNA?
(iii) Which makes proteins?
(iv) Which digests waste?

END OF QUESTION PAPER Total: 40 Marks

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